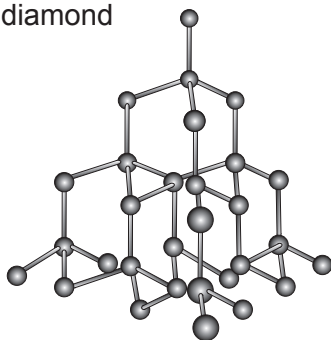


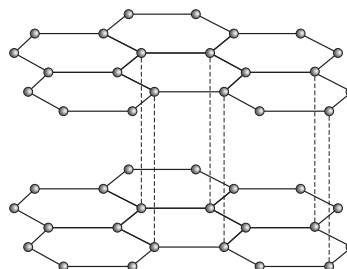
7. Allotropes of carbon include diamond, graphite, graphene, carbon fibres, fullerenes and carbon nanotubes.

(a) The structures of diamond and graphite are shown below.

diamond



graphite



Answer the following questions with reference to the bonding in diamond and graphite.

- (i) Explain why diamond is an insulator of electricity but graphite is a conductor. [4]

.....

.....

.....

.....

- (ii) Explain why graphite is soft and slippery but diamond is hard. [3]

.....

.....

.....

.....